

A Distributed Voltage Controller for Medium Voltage Grids with Storage-containing Loads

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Abstract—In this paper, a set-up consisting of a medium voltage grid with connected water distribution network as a load is investigated. The water distribution network contains water reservoirs that can be used as storages to support the voltage control by adjusting the electric power consumption to keep the voltages in the required limitations. At the same time the electric power and water demand are satisfied. The considered set-up is derived from a real-world set-up in the city of Kaiserslautern, Germany, where the power grid and water distribution grid are operated by the same operator. A distributed model predictive controller for the voltage control is designed in general form and thus the concept be applied to other loads containing storages. The controller is tested in simulations.

I. INTRODUCTION

Driven by the intention for an environmental friendly electricity production, the electric power grid is currently undergoing large changes. More and more small generation units, mostly renewable, are installed at low or medium voltage (LV or MV) level. As they are usually operated uncontrolled, large voltages increases at the nodes or currents in the branches can occur. To coordinate the in-feed and match them to the loads, the application of modern control methods is required. One possibility to alter the voltage is to provide additionally reactive power by the distributed generation units as they are usually connected to the grids via inverters. Another option is to change the power demand of the system. This is especially effective if the loads have a high power consumption and local storages nearby so that the process inside the load does not have to be interrupted. The storages can be charged at times when the load is low and discharged if there is a peak demand.

In this paper, a real-world example of a radial medium voltage grid where one of the nodes is coupled to a water distribution network (WDN) is investigated. More specifically, at one of the nodes a pump is installed which primal purpose is to transport water in the WDNs in order to satisfy the consumer demand. The WDN is also equipped with a water reservoir in which water can be stored for the satisfaction of the demand at peak times and for emergency cases such as fire or blackouts in the power grids. Due to hygienic reasons, the storages have to be emptied and refilled over the day. This is usually done at a specific time of the day

without considering the current status of the power grid. The main idea of the control approach is to exploit the storage in the WDN to help to control the voltage in the power system in combination with the control of regenerative generation units. The actual set-up is derived from the power grid and the WDN of the city of Kaiserslautern, Germany, where both systems are operated independently by the same operator. This is a typical situation in many cities where the interactions between the grids are not considered although a common operation is beneficial in terms of performance improvement in the power grid. In this paper it is discussed how the potential can be exploited using modern control methods.

As power systems and WDNs are usually subject to constraints, such as physical limitations, model predictive control (MPC) is a reasonable control method. In case of MV power systems, maintaining the voltage of the respective feeders inside a given band is one of the main operational challenges. Especially the use of model based and predictive control methods to provide voltage support as an ancillary service has found considerable attention. In [1] and [2] a combined approach is introduced that relies on a local primary control with PI-controllers and a centralized upper layer MPC based secondary control structure. The MPC calculates the optimal set-points considering system limitations while the local controllers react to faster disturbances in real-time. A direct operation of a MV grid with MPC is presented in [3]. Besides grid wide volt/var control, the authors investigate the coupling between high voltage (HV) and MV.

Since power systems and WDNs are often large scale, it is often desired to decompose the overall system into several subsystems which are then controlled by individual MPCs. To coordinate the decision making, the controllers communicate with each other. This methodology is also known as distributed model predictive control (DMPC). More information about DMPC can for instance be found in the survey paper [4]. Interestingly, while the joined coordination of gas and electricity [5] as well as the control of WDNs with DMPC has been studied in literature, see e.g. [6], [7], investigations on voltage support with WDNs in MV grids using DMPC are rare. This is probably the case since simple but accurate models for the power flow calculation are missing in the literature. These models are however required for efficient calculations of the optimization problems at the core of the DMPC problems in real-time. For this purpose, a linear power flow approximation for radial low voltage grids in combination with a privacy preserving DMPC controller is proposed in our previous work [8]. The results presented

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here extend them in the following directions:

1) Set-up: A radial medium voltage grid in combination with a WDN is investigated. The main idea is to cooperatively control voltage by adjusting the power in-feed from the regenerative generation units in combination with exploiting the water storage in the WDNs. At the same time the demands in the electric power system as well as the WDNs are covered.

2) Distributed Controller: A cooperative, iterative DMPC controller is designed. Due to the periodicity in the demands of the power system and the WDN, the controller is periodic. With the inclusion of soft constraints for the voltage limitations, recursive feasibility and convergence to a periodic trajectory can be shown. To reduce the communication effort, the controller uses event-based communication, i.e., subsystems only exchange information with each other when there is a necessity.

The remainder of the paper is structured as follows: After introducing the set-up in Section II, the modeling of the power network is presented in Section III. The controller design is discussed in Section IV. Section V contains the simulation results and Section VI concludes the paper.

Notation $\mathcal{I}_{n_1:n_2}$ is the set of integers from n_1 to n_2 . For two integers n and $m \neq 0$, $[n]_m$ is the modulo operation. $\underline{v}$ denotes a complex value, i.e., $\underline{v} \in \mathbb{C}$. Given a complex value $\underline{v}$, $\Re\{\underline{v}\}$, $\Im\{\underline{v}\}$, $|\underline{v}|$ and $\bar{\underline{v}}$ denote its real part, imaginary part, absolute value and conjugate complex value, respectively. Given a vector $\mathbf{v} \in \mathbb{R}^n$, $\|\mathbf{v}\|$ denotes the Euclidean norm. $\emptyset$ is the empty set. For vectors $\mathbf{x}_i \in \mathbb{R}^{n_i}$ for $i \in \mathcal{N} \subset \mathbb{N}$, where $\mathcal{N}$ is ordered, the vector $\mathbf{x}_{\mathcal{N}} = \text{col}_{i \in \mathcal{N}}(\mathbf{x}_i) \in \mathbb{R}^n$ with $n = \sum_{i \in \mathcal{N}} n_i$ is given by $[\mathbf{x}_{\mathcal{N}(1)}^T, \dots, \mathbf{x}_{\mathcal{N}(|\mathcal{N}|)}^T]^T$ where $\mathcal{N}(i)$ is the i -th element and $|\mathcal{N}|$ the cardinality of $\mathcal{N}$. $\mathbf{D} = \text{diag}_{i \in \mathcal{N}}[\mathbf{D}_i]$ denotes a block-diagonal matrix with matrices $\mathbf{D}_i$ on its diagonal.

II. PROBLEM SET-UP

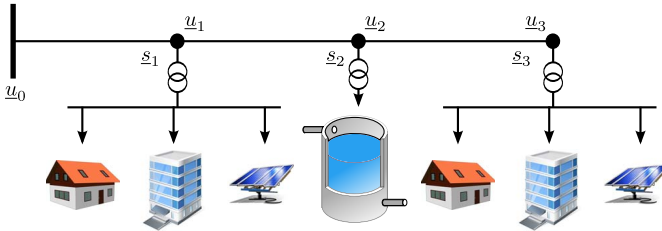


Fig. 1: Structure of considered radial MV power grid with WDN connected at node 2

In the paper a radial medium voltage grid with $N_N = 3$ nodes as shown in Fig. 1 is considered. The set-up is derived from a real-world example. In the first and the third node there are aggregated consumers (households and companies) which draw power from the grid and solar plants which injects power into the grid. The solar plants are connected to the grid via inverters and so active and reactive power can be controlled. The demand from the consumers is assumed to be uncontrollable. In the second node a pump station is

installed. The major purpose of the pump is to supply water to the consumers of the drinking water distribution network (WDN). The WDN comprises a water reservoir in which water can be stored. The reservoirs are required to buffer the water for times with a high water consumption or the case of blackouts in the power grid or fires. In this paper it is investigated how the water reservoir can be used for the control of the voltage in the grid. It is assumed that the owner of the WDN and the power grid are identical and so have an interest to utilize the WDN for ancillary services in the power grid.

The objective of the paper is to design a controller for the considered power grid where the solar panels and the water reservoir in the WDN are exploited to regulate to keep voltage within its bounds. Although the power grid considered in the paper is rather small, the controller has to be implementable in a distributed fashion such that an application to grids consisting of more nodes is possible.

III. MODELING

A. Power Grid

In this section a linear approximation for the power flow in the network is reviewed. It is originally presented in [8]. All powers drawn from/fed into the grid are by definition negative/positive. Based on Z- and P-load models, the currents $\underline{i}_\nu$ and voltages $\underline{u}_\nu$ in the node ν of the radial grid can be described by

$$\underline{i}_\nu = -\frac{\underline{u}_\nu}{\underline{z}_\nu} + \frac{\bar{\underline{s}}_\nu}{\underline{u}_\nu}, \quad (1a)$$

$$\underline{u}_\nu = \mathbf{1}_\nu^T \underline{\mathbf{Z}} \mathbf{i} + \underline{u}_0 \quad (1b)$$

for all $\nu \in \mathcal{I}_{1:N_N}$. Furthermore, $\underline{z}_\nu$ the load impedance and $\underline{s}_\nu$ the apparent power in the node. $\underline{u}_0$ is the voltage in the slack bus and $\mathbf{i} = \text{col}_{\nu \in \mathcal{I}_{1:N_N}}(\underline{i}_\nu)$. $\underline{\mathbf{Z}}$ is the constant impedance matrix of the grid. The vector $\mathbf{1}_\nu^T = [0 \dots 0 \ 1 \ 0 \dots 0]$ maps the corresponding voltage drops across the power line from the slack bus to node ν . For given values $\underline{u}_0$, $\underline{z}_\nu$ and $\bar{\underline{s}}_\nu$, the nonlinear power flow equations (1) can be solved for the voltages $\underline{u}_\nu$ and currents $\underline{i}_\nu$ for all ν . For this purpose for instance the Forward-Backward-Sweep (FBS) method can be applied.

For the controller implementation, a linear relation between current, voltage and power is appealing. Substituting (1b) into (1a) for all $\nu \in \mathcal{I}_{1:N_N}$, linearizing the resulting equations around an operating point and separating them into real and imaginary part one obtains the linear relation

$$\mathbf{0} = \mathbf{A}_{\text{PF}} \cdot \mathbf{i} + \mathbf{B}_{\text{PF}} \cdot \mathbf{s} + \mathbf{c}_{\text{PF}} \quad (2a)$$

$$\mathbf{u} = \mathbf{C}_{\text{PF}} \cdot \mathbf{i} + \mathbf{u}_0, \quad (2b)$$

$$\text{with } \mathbf{u}_0 = \begin{bmatrix} 1 \cdot \Re\{\underline{u}_0\} \\ 1 \cdot \Im\{\underline{u}_0\} \end{bmatrix}, \mathbf{u} = \begin{bmatrix} \mathbf{u}^{\text{Re}} \\ \mathbf{u}^{\text{Im}} \end{bmatrix}, \mathbf{s} = \begin{bmatrix} \mathbf{p} \\ \mathbf{q} \end{bmatrix}, \mathbf{i} = \begin{bmatrix} \mathbf{i}^{\text{Re}} \\ \mathbf{i}^{\text{Im}} \end{bmatrix}$$

where $\mathbf{i}^{\text{Re}} = \text{col}_{\nu \in \mathcal{I}_{1:N_N}}(\Re(\underline{i}_\nu))$ and $\mathbf{i}^{\text{Im}} = \text{col}_{\nu \in \mathcal{I}_{1:N_N}}(\Im(\underline{i}_\nu))$ as well as $\mathbf{u}^{\text{Re}} = \text{col}_{\nu \in \mathcal{I}_{1:N_N}}(\Re(\underline{u}_\nu))$ and $\mathbf{u}^{\text{Im}} = \text{col}_{\nu \in \mathcal{I}_{1:N_N}}(\Im(\underline{u}_\nu))$ are the stacked real and imaginary parts of the node currents as well as

voltages, respectively. $\mathbf{p} = \text{col}_{\nu \in \mathcal{I}_{1:N_N}}(\Re(\underline{s}_\nu))$ and $\mathbf{q} = \text{col}_{\nu \in \mathcal{I}_{1:N_N}}(\Im(\underline{s}_\nu))$ are the stacked active and reactive node powers. For the derivation and the computation of the constant matrices $\mathbf{A}_{\text{PF}} \in \mathbb{R}^{2N_N \times 2N_N}$, $\mathbf{B}_{\text{PF}} \in \mathbb{R}^{2N_N \times 2N_N}$ and $\mathbf{C}_{\text{PF}} \in \mathbb{R}^{2N_N \times 2N_N}$ please refer to [8]. The matrix $\mathbf{A}_{\text{PF}}$ has full rank. Note that different from [8] where LV grids are considered, this paper deals with the MV grids where the capacitance of the cables needs to be included. This is considered in the additional term $\mathbf{c}_{\text{PF}} \in \mathbb{R}^{2N_N \times 1}$. Moreover, the local stacked voltages $\mathbf{u}_\nu^T = [\Re\{\underline{u}_\nu\} \ \Im\{\underline{u}_\nu\}]$, currents $\mathbf{i}_\nu^T = [\Re\{\underline{i}_\nu\} \ \Im\{\underline{i}_\nu\}]$ and apparent powers $\mathbf{s}_\nu^T = [\Re\{\underline{s}_\nu\} \ \Im\{\underline{s}_\nu\}] = [p_\nu \ q_\nu]$ are introduced. Notably, this linear formulation has some appealing properties such as its decomposability and explicit formulation. Furthermore, in contrast to other approximations found in literature it can be adapted to the operating point of the grid.

To allow a safe operation of the power grid, the nodal voltage $\underline{u}_\nu$ have to be bounded. Therefore, the set

$$\tilde{\mathcal{U}}_\nu = \left\{ \mathbf{u}_\nu \in \mathbb{R}^2 \mid u_{\min}^2 \leq \|\mathbf{u}_\nu\|^2 \leq u_{\max}^2 \right\}. \quad (3)$$

is defined. Note that the set is non-convex. To obtain a computational more attractive set, it is approximated by the convex set

$$\mathcal{U}_\nu = \left\{ \begin{bmatrix} u_\nu^1 \\ u_\nu^2 \end{bmatrix} \in \mathbb{R}^2 \mid \left\| \begin{bmatrix} u_\nu^1 \\ u_\nu^2 \end{bmatrix} \right\|^2 \leq u_{\max}^2 \wedge u_\nu^1 \geq u_{\min} \right\}.$$

Note that the approximation is sufficiently accurate as the voltage angle does not vary strongly for the considered type of MV grids. The constraints (3) on the voltages are no hard constraints so in the optimization problem the soft constraints

$$\mathcal{U}_\nu^s = \left\{ \begin{bmatrix} u_\nu^1 \\ u_\nu^2 \end{bmatrix} \in \mathbb{R}^2, \epsilon_\nu^m, \epsilon_\nu^M \mid \left\| \begin{bmatrix} u_\nu^1 \\ u_\nu^2 \end{bmatrix} \right\|^2 \leq u_{\max}^2 + \epsilon_\nu^M \wedge u_\nu^1 \geq u_{\min} - \epsilon_\nu^m \right\} \quad (4)$$

is integrated. In combination with weighting the soft constraint variables $\epsilon_\nu^m \geq 0$, $\epsilon_\nu^M \geq 0$, a violation of the constraints is possible but at a high cost, which is why the controller tries to avoid it.

As the power in-feed or load of some components cannot be adjusted by the controller, the apparent node power is split into an fixed and a variable part, i.e., $\mathbf{s}_\nu = \mathbf{s}_{f,\nu} + \mathbf{s}_{v,\nu}$. In the nodes 1 and 3, the power of the consumers cannot be influenced and thus they belong to the fixed part $\mathbf{s}_{f,\nu} = \mathbf{s}_{C,\nu}$. They are modeled through given trajectories which can be obtained from load profiles of the consumers. More information is given in Section V.

The variable power part $\mathbf{s}_{v,\nu} = \mathbf{s}_{\text{PV},\nu} \ \forall \nu \in \{1, 3\}$ stems from the in-feed from the solar plants. Here the active power $p_{\text{PV},\nu}$ is assumed to be uncontrollable and is taken from the prediction of the in-feed. The reactive power of the solar plant $q_{\text{PV},\nu}$ can be controlled and thus used to control the voltage in the grid. The apparent power of the inverters is limited $\forall \nu \in \{1, 3\}$ by

$$\mathbf{s}_{v,\nu} \in \mathcal{S}_\nu = \left\{ \mathbf{s} = \begin{bmatrix} p_{\text{PV},\nu} \\ q_{\text{PV},\nu} \end{bmatrix} \mid p_{\text{PV},\nu}^2 + q_{\text{PV},\nu}^2 \leq s_{\max,\nu}^2 \right\}, \quad (5)$$

where $s_{\max,\nu}$ is the maximum allowed apparent power.

B. Water Distribution Network

The water level h_t in the reservoir of the WDN can be described by

$$h_{t+1} = h_t + \frac{T_s}{A_B} \cdot (q_t - d_{W,t}), \quad (6)$$

$\forall t \in \mathbb{N}$ where T_s is the sample time, A_B the base area of the water tank, $q_t \geq 0$ is the water flow into the reservoir and $d_{W,t}$ represents the water demand at the time step t . The water flow q_t can be controlled by the pump. Thereby, active power is drawn from the grid. The relation is linearly approximated by $q_t = c_p \cdot p_{2,t}$ where c_p is a constant which can be determined from the pump characteristics provided by the pump manufacturer. The water demand $d_{W,t}$ is uncontrollable. As the water demand does not change significantly from day to day, it can be approximated with measurement data from the WDN operator. The water reservoir has a limited capacity and is thus subject to the constraints $h_{\min} \leq h_t \leq h_{\max}$. Similar to before the power drawn from the pump is limited by

$$\mathbf{s}_{v,2} \in \mathcal{S}_2 = \left\{ \mathbf{s}_2 = \begin{bmatrix} p_2 \\ q_2 \end{bmatrix} \mid p_2^2 + q_2^2 \leq s_{\max,2}^2 \right\}, \quad (7)$$

as the pump is connected to the grid through an inverter. The reactive power is controlled to zero $q_2 = 0$. There is no fixed part in node 2, thus $\mathbf{s}_{f,2} = \mathbf{0}$ for all $t \in \mathbb{N}$.

C. Overall Model

In the sequel the more general representation of (6)

$$\mathbf{x}_{t+1} = \mathbf{A}\mathbf{x}_t + \mathbf{B}\mathbf{s}_{v,2,t} + \mathbf{B}_d\mathbf{d}_t, \quad (8)$$

$\forall t \in \mathbb{N}$ with the state $\mathbf{x}_t = \mathbf{h}_t$, known disturbance $\mathbf{d}_t = d_{W,t}$, system matrix $\mathbf{A} = 1$, input matrix $\mathbf{B} = -\frac{T_s}{A_B} \cdot c_B [1 \ 0]$ and $\mathbf{B}_d = \frac{T_s}{A_B}$ is used. The water level is subject to the constraints $\mathbf{x}_t \in \mathcal{X} \ \forall t \in \mathbb{N}$ where $\mathbf{x}_t \in \mathcal{X} = \{\mathbf{x} \mid x_{\min} \leq \mathbf{x} \leq x_{\max}\}$ with $x_{\min} = h_{\min}$ and $x_{\max} = h_{\max}$.

Moreover, the global apparent power set $\mathcal{S} = \prod_{\nu=1}^{N_N} \mathcal{S}_\nu$ is introduced. A direct relationship between apparent powers and the voltages is obtained by substituting (2a) in (2b), i.e., $\mathbf{u} = \mathbf{D}_{\text{PF}}\mathbf{s} + \mathbf{d}_{\text{PF}}$ where $\mathbf{D}_{\text{PF}} = -\mathbf{C}_{\text{PF}}\mathbf{A}_{\text{PF}}^{-1}\mathbf{B}_{\text{PF}}$ and $\mathbf{d}_{\text{PF}} = -\mathbf{C}_{\text{PF}}\mathbf{A}_{\text{PF}}^{-1}\mathbf{c}_{\text{PF}} + \mathbf{u}_0$. Note that $\mathbf{A}_{\text{PF}}$ has full rank. Introducing the global voltage set $\mathcal{U}^s = \prod_{\nu=1}^M \mathcal{U}_\nu^s$ and the global soft constraint variable $\boldsymbol{\epsilon} = \text{col}_{i \in \mathcal{I}_{1:N_N}} \left(\begin{bmatrix} \epsilon_\nu^m \\ \epsilon_\nu^M \end{bmatrix} \right)$, the voltage soft constraint (4) can be recast as

$$(\mathbf{D}_{\text{PF}}\mathbf{s} + \mathbf{d}_{\text{PF}}, \boldsymbol{\epsilon}) \in \mathcal{U}^s, \quad \boldsymbol{\epsilon} \geq \mathbf{0}.$$

IV. DISTRIBUTED CONTROLLER

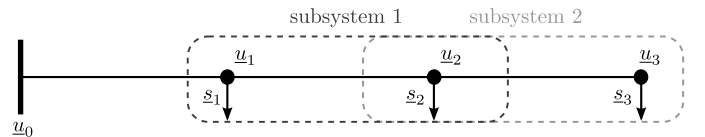


Fig. 2: Structure of overlapping subsystems

This section discusses the distributed control scheme. To allow cooperation $M = 2$ overlapping subsystems as shown

in Fig. 2 are introduced. The set Σ_i contains the indexes of the node belonging to the subsystem i . For the considered set-up this means that $\Sigma_1 = \{1, 2\}$ and $\Sigma_2 = \{2, 3\}$. Moreover, the set $\bar{\Sigma}_i = \mathcal{I}_{1:M} \setminus \Sigma_i$ is defined. It is assumed that the in-feed power from solar panels and the power consumption as well as the water demand do not change significantly over the days. In other words, it is assumed that they are periodic with periodicity P , i.e., $p_{PV,\nu,t} = p_{PV,\nu,t+P}$, $s_{f,\nu,t} = s_{f,\nu,t+P}$ and $d_{W,t} = d_{W,t+P}$ for all $t \in \mathbb{N}$. Due to this a periodic controller is designed. It makes use of an optimal trajectory which is obtained by solving the problem

$$\underset{\bar{\mathbf{X}}, \bar{\mathbf{S}}_v, \bar{\mathbf{E}}}{\text{minimize}} \quad \sum_{r=0}^{P-1} l(\bar{\mathbf{x}}_r, \bar{\mathbf{s}}_{v,r}, \bar{\mathbf{e}}_r) \quad (9a)$$

$$\text{subject to} \quad \bar{\mathbf{x}}_0 = \bar{\mathbf{x}}_P \quad (9b)$$

for all $r \in \mathcal{I}_{0:P-1}$:

$$\bar{\mathbf{x}}_{r+1} = \mathbf{A}\bar{\mathbf{x}}_r + \mathbf{B}\bar{\mathbf{s}}_{a2,r} + \mathbf{B}_d \mathbf{d}_r \quad (9c)$$

$$\bar{\mathbf{x}}_r \in \mathcal{X} \quad (9d)$$

$$\bar{\mathbf{s}}_{v,r} \in \mathcal{S} \quad (9e)$$

$$(\mathbf{D}_{PF}(\bar{\mathbf{s}}_{v,r} + \mathbf{s}_{f,r}) + \mathbf{d}_{PF}, \bar{\mathbf{e}}_r) \in \mathcal{U}^s \quad (9f)$$

$$\bar{\mathbf{e}}_r \geq \mathbf{0}. \quad (9g)$$

The optimal trajectories are denoted with $*$. Constraint (9b) ensures that the solution is periodic. The remaining constraints are taken from the overall model in Section III-C. The stage cost is $l(\mathbf{x}_r, \mathbf{s}_{u,r}, \mathbf{e}_r) = \tilde{\mathbf{x}}_r^T \mathbf{Q}_x \tilde{\mathbf{x}}_r + \mathbf{s}_{v,r}^T \mathbf{Q}_S \mathbf{s}_{v,r} + \mathbf{e}_r^T \mathbf{Q}_e \mathbf{e}_r$ where $\tilde{\mathbf{x}}_l = \mathbf{x}_l - \mathbf{x}_{\text{ref}}$. $\mathbf{x}_{\text{ref}}$ is a reference value for the state and $\mathbf{Q}_x$, $\mathbf{Q}_S$ and $\mathbf{Q}_e$ are positive definite weighting matrices. Optimization problem (9) can be recast as a strictly convex quadratically constrained quadratic program (QCQP).

The DMPC algorithm used in the paper works iteratively and is shown in Algorithm 1. Therein, the following optimization problem is solved in each subsystem

$$\underset{\mathbf{X}, \mathbf{S}_v, \mathbf{E}}{\text{minimize}} \quad V_N(\bar{\mathbf{X}}_t, \bar{\mathbf{S}}_{v,t}, \bar{\mathbf{E}}_t) = \sum_{r=t}^{t+N-1} l(\bar{\mathbf{x}}_r, \bar{\mathbf{s}}_{v,r}, \bar{\mathbf{e}}_r) \quad (10a)$$

$$\text{subject to} \quad \mathbf{x}_t = \mathbf{x}_t^m \quad (10b)$$

$$\mathbf{x}_{t+N} = \bar{\mathbf{x}}_{[t+N]P}^* \quad (10c)$$

for all $r \in \mathcal{I}_{t:t+N-1}$:

$$\mathbf{x}_{r+1} = \mathbf{A}\mathbf{x}_r + \mathbf{B}\mathbf{s}_{v,2,r} + \mathbf{B}_d \mathbf{d}_r \quad (10d)$$

$$\mathbf{x}_r \in \mathcal{X} \quad (10e)$$

$$\mathbf{s}_{v,\nu,r} \in \mathcal{S}_\nu \quad \forall \nu \in \Sigma_i \quad (10f)$$

$$\mathbf{s}_{v,\nu,r} = \hat{\mathbf{s}}_{v,\nu,r}^k \quad \forall \nu \in \bar{\Sigma}_i \quad (10g)$$

$$(\mathbf{D}_{PF}(\mathbf{s}_{v,r} + \mathbf{s}_{f,r}) + \mathbf{d}_{PF}, \mathbf{e}_r) \in \mathcal{U}^s \quad (10h)$$

$$\mathbf{e}_r \geq \mathbf{0} \quad (10i)$$

where $\bar{\mathbf{X}}_t = [\bar{\mathbf{x}}_t^T, \dots, \bar{\mathbf{x}}_{t+N}^T]^T$, $\bar{\mathbf{S}}_v = [\bar{\mathbf{s}}_{v,t}^T, \dots, \bar{\mathbf{s}}_{v,t+N-1}^T]^T$ and $\bar{\mathbf{E}} = [\bar{\mathbf{e}}_t^T, \dots, \bar{\mathbf{e}}_{t+N-1}^T]^T$ are predicted sequences of the state, apparent power and soft constraint variable, respectively. N is the prediction horizon. In (10b) the measurement of the state $\mathbf{x}_t^m$ is included into the optimization problem. The constraint (10b) ensures that the terminal set lies on the optimal trajectory. With its help recursive feasibility can be

Algorithm 1 DMPC algorithm

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1: Measure  $\mathbf{x}_t^m$ 
2: for  $k = 1, \dots, k_{\max}$ 
3:   Each Subsystem  $i$ :
       Solve (10) to obtain  $[\mathbf{X}_t^{[i],k+1}, \mathbf{S}_{v,t}^{[i],k+1}, \mathbf{E}_t^{[i],k+1}] =$ 
          $\text{argmin}_{\mathbf{X}, \mathbf{S}_v, \mathbf{E}} V_N(\mathbf{X}, \mathbf{S}_v, \mathbf{E})$ 
       Compute  $\Delta V_N^{[i]} = V_N(\mathbf{X}_t^{[i],k+1}, \mathbf{S}_{v,t}^{[i],k+1}, \mathbf{E}_t^{[i],k+1}) -$ 
          $V_N(\hat{\mathbf{X}}_t^k, \hat{\mathbf{S}}_{v,t}^k, \hat{\mathbf{E}}_t^k)$ 
       Evaluate event-trigger  $\delta_i = \begin{cases} 1, & \text{if } \Delta V_N^{[i]} < \gamma \\ 0, & \text{else} \end{cases}$ 
4: Find  $i^* = \text{argmin}_{i \in \mathcal{D}} \Delta V_N^{[i]}$  where  $\mathcal{D} = \{i \in \mathcal{I}_{1:M} | \delta_i = 1\}$ 
5: if  $\mathcal{D} = \emptyset$ 
       Set  $\hat{\mathbf{S}}_{v,t}^{k+1} = \hat{\mathbf{S}}_{v,t}^k$ ,  $\hat{\mathbf{X}}_t^{k+1} = \hat{\mathbf{X}}_t^k$  and  $\hat{\mathbf{E}}_t^{k+1} = \hat{\mathbf{E}}_t^k$ 
6: else
        $i^*$  sends  $\mathbf{S}_{v,t}^{[i^*],k+1}$ ,  $\mathbf{X}_t^{[i^*],k+1}$ ,  $\mathbf{E}_t^{[i^*],k+1}$  to all subsystems
       Set  $\hat{\mathbf{S}}_{v,t}^{k+1} = \mathbf{S}_{v,t}^{[i^*],k+1}$ 
       Set  $\hat{\mathbf{X}}_t^{k+1} = \mathbf{X}_t^{[i^*],k+1}$ 
       Set  $\hat{\mathbf{E}}_t^{k+1} = \mathbf{E}_t^{[i^*],k+1}$ 
7: if  $\mathcal{D} = \emptyset$  or  $k = k_{\max}$ 
       Set  $\hat{\mathbf{S}}_{v,t+1}^1 = [\hat{\mathbf{s}}_{v,t}^{k+1,T}, \dots, \hat{\mathbf{s}}_{v,t+N-1}^{k+1,T}, \bar{\mathbf{s}}_{v,[t+N]P}^{*,T}]^T$ ,
       Set  $\hat{\mathbf{X}}_{t+1}^1 = [\hat{\mathbf{x}}_t^{k+1}, \dots, \hat{\mathbf{x}}_{t+N}^{k+1}, \bar{\mathbf{x}}_{[t+N]P}^{*,T}]^T$ ,
       Set  $\hat{\mathbf{E}}_{t+1}^1 = [\hat{\mathbf{e}}_t^{k+1}, \dots, \hat{\mathbf{e}}_{t+N-1}^{k+1}, \bar{\mathbf{e}}_{[t+N]P}^{*,T}]^T$ ,
       break for loop
     end for
8: Apply  $\mathbf{s}_{v,t} = \hat{\mathbf{s}}_{v,t}^{k+1}$  to the plant

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ensured, see [9] for more information. Constraint (10f+g) ensures that the optimization is only performed over the inputs of the nodes included in the subsystem, i.e., $\mathbf{s}_{v,\nu,r}^k$ with $\nu \in \Sigma_i$ and for the remaining subsystems the exchanged trajectories $\bar{\mathbf{s}}_{a,\nu,r}$ are used. k is the iteration counter of the algorithm. The local problems can be recast as strictly convex QCQPs. This implies that the state $\mathbf{x}_t$ converges to the optimal trajectory $\bar{\mathbf{X}}^*$, see again [9] for more information.

In step 3 the optimization problem (10) is solved in every subsystems and the cost decrease ΔV_N to the previous solution is computed. To distinguish between the solution of the different subsystems, they are indexed with $[i]$. If the cost decrease is larger than a predefined threshold $\gamma \leq 0$, then an event is triggered (by setting the binary variables δ_i to one). This is done to avoid unnecessary iterations between the subsystems in the case no or only a small progress is obtained. From the subsystems which triggered an event, the one with the maximum cost decrease is selected and named i^* (step 4). If a subsystem i^* is found, then the solution is sent to the remaining subsystems and the trajectories are included in the optimization problem for the next iteration (step 6). Otherwise the last feasible solution is maintained (step 5). The procedure is repeated until the maximum number of iterations $k_{\max}$ is reached or the no sufficient large cost decrease is made. In step 7, initially feasible solutions for first iteration in the next time step are generated by extending the feasible solution found with the suitable part of the optimal trajectory. As standard in MPC, the first input is given to the plant in step 8.

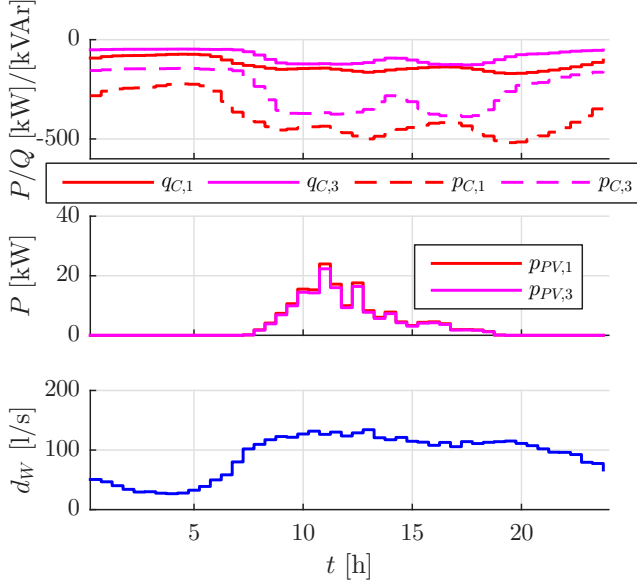


Fig. 3: Load power, in-feed from solar plants and water consumption over one day

For $t = 0$ initially feasible sequences $\hat{\mathbf{X}}_0^1, \hat{\mathbf{S}}_{v,0}^1, \hat{\mathbf{E}}_0^1$ have to be found. This is done by setting $p_{2,r} = \frac{1}{c_B} d_{W,r}, q_{1,r} = q_{3,r} = 0 \forall r \in \mathcal{I}_{0:N_N}$ and $\hat{\mathbf{x}}_0^1 = \mathbf{x}_0^m$. $\hat{\mathbf{E}}_0^1$ is then chosen such that the soft constraints hold. The approach generates feasible solutions as all constraints in (10) are then satisfied.

V. SIMULATION

A. Set-up

The grid data can be found in Table I. The minimum and maximum voltage are chosen $\pm 2\%$ of the nominal voltage.

In the top figure of Fig. 3, the active and reactive consumer power $p_{C,\nu}$ and $q_{C,\nu}$ are shown. In the figure in the middle of Fig. 3 the active power in-feed $p_{PV,\nu}$ for $\nu \in \{1, 3\}$ from the solar plants are depicted. The in-feeds are almost the same as the solar plants are located close to each other but have a different peak power. The power in-feed is much lower than the peak power as the data stems from a winter day. In the bottom figure, the water demand d_W is shown. All data is taken from measurements of a real-world MV grid or WDN. The in-feed and consumer powers as well as the water demand are assumed to be periodic, i.e., are the same for every day. The weighting matrices are $\mathbf{Q}_x = 0.01, \mathbf{Q}_s = \text{diag}_{\nu \in \mathcal{I}_{1:N_N}} [\mathbf{Q}_{S\nu}], \mathbf{Q}_\epsilon = 10^6 \cdot \mathbf{I}$ with $\mathbf{Q}_{S1} = \mathbf{Q}_{S2} = 1 \cdot \mathbf{I}, \mathbf{Q}_{S3} = 10 \cdot \mathbf{I}$. Note that the cost for reactive power in node 3 is higher so that the input of node 1 is preferred. An in-feed of node 3 increases the losses in the branches because of the larger distance and as a result the node voltages sink. The matrix $\mathbf{Q}_\epsilon$ weights the penalization of the violation of the voltage constraints. The weight is chosen relatively high so that a violation is avoided.

The communication between the controllers can be internet-based. Due to the large sampling time, communication delays are not considered. All optimization problems

TABLE I: simulation parameters

description	name	value
line resistance	R'	0.264Ω
line inductance	X'_L	0.189 mH
line capacitance	X'_C	$0.27 \mu\text{F}$
line length slack bus to node 1	l_{01}	8 km
line length node 1 to 2	l_{12}	4 km
line length node 2 to 3	l_{23}	4 km
base area	A_B	833.3 m^2
max. tank level	$h_{\max}$	5.65 m
min. tank level	$h_{\min}$	3 m
power factor	c_B	$-519.73 \frac{1}{\text{Ws}}$
sample time	T_s	30 min
periodicity	P	48
prediction horizon	N	24
max. iterations	$k_{\max}$	10
max. apparent power	$s_{\max,1}$	250 kVA
max. apparent power	$s_{\max,2}$	118.5 kVA
max. apparent power	$s_{\max,3}$	233 kVA
max. voltage	$u_{\max}$	11.78 kV
min. voltage	$u_{\min}$	11.32 kV

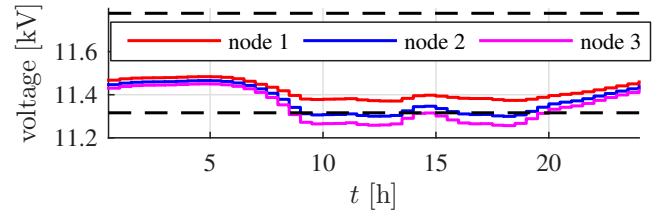


Fig. 4: Uncontrolled nodal voltages

are solved with CPLEX 12.6 on a Windows 7 PC with an Intel Core i5-660 processor with 3.33 GHz and 8 GB RAM.

B. Simulation Results

In Fig. 4 the uncontrolled nodal voltages are shown. Obviously, a controller is required to prevent the voltages in node 2 and 3 to violate the lower voltage limitation.

The simulation results of with controller are shown in the Fig. 5. It can be observed that the controller respects all the constraints on the voltage, power and water level. Moreover, the water level converges to the optimal trajectory h^* obtained by solving (9) offline. Considering the inputs of active- and reactive power it can be seen that the water tank is filled over the day when the sun is shining and the PV systems inject active power in the system. Parallel to the injection of active power, reactive power is injected in the nodes 1 & 3, whereas node 1 injects more reactive power due to the higher cost in node 3. The input of active- and reactive power prevents that the nodal voltages break through their lower boundaries. Note that the tank is filled before and after the peak load times when the voltage are not close to the boundaries. The lower subfigure shows the updating subsystem of the DMPC controller over the iterations. At the start of the simulation DMPC algorithm iterates to find a solution, when the optimal trajectory is reached no more

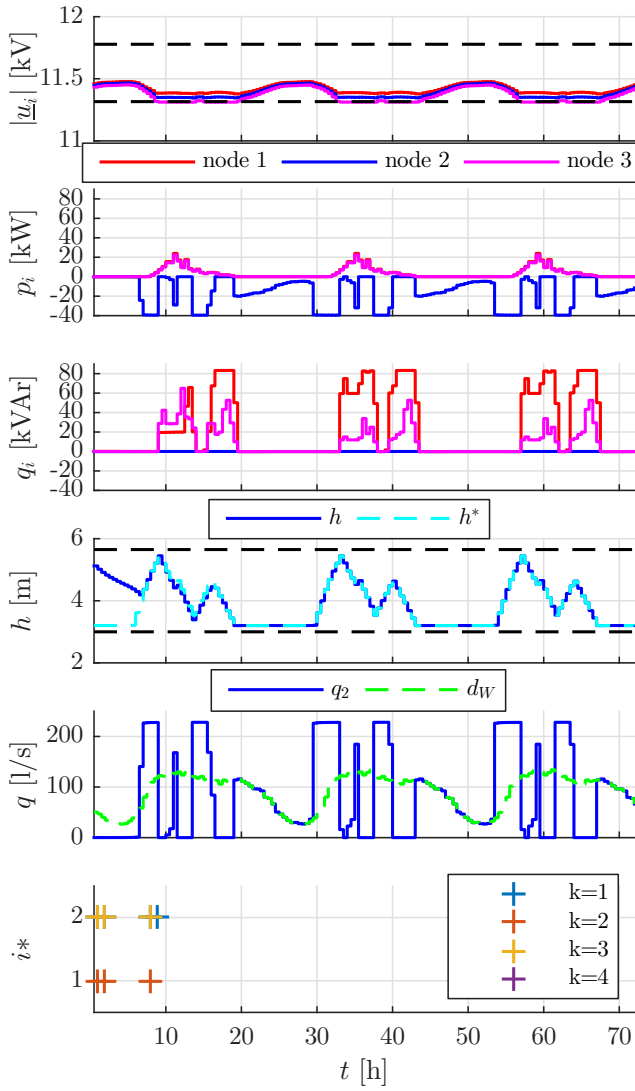


Fig. 5: Simulation results with DMPC

iterations are required.

The DMPC is also compared to a centralized MPC (CMPC) with full model information. The closed-loop cost $J = \sum_{t=0}^{T_{\text{sim}}} l(\mathbf{x}_t, \mathbf{s}_{u,t}, \epsilon_t)$ are $J_{\text{CMPC}} = 0.0516$ pu and $J_{\text{DMPC}} = 0.1154$ pu showing the DMPC is performing worse due to the distributed decision making and the incomplete knowledge in the subsystems. Considering the mean computational time of both controllers $t_{\text{CMPC}} = 0.2268$ s and $t_{\text{DMPC}} = 0.1332$ s, it can be seen that the CMPC controller needs in average $\approx 40\%$ more time than the DMPC.

Fig. 6 shows the results when the controller is applied to a more realistic set-up (also called set-up 2) where the power from consumers and the in-feed from solar plants is not perfectly periodic but varies over the day and the water consumption is different from the predicted one. To prevent the controller from becoming infeasible, the additional tightened soft constraints on the water level are used and the terminal constraint (10c) is removed. One can see that the controller respects the voltage constraints most of the time.

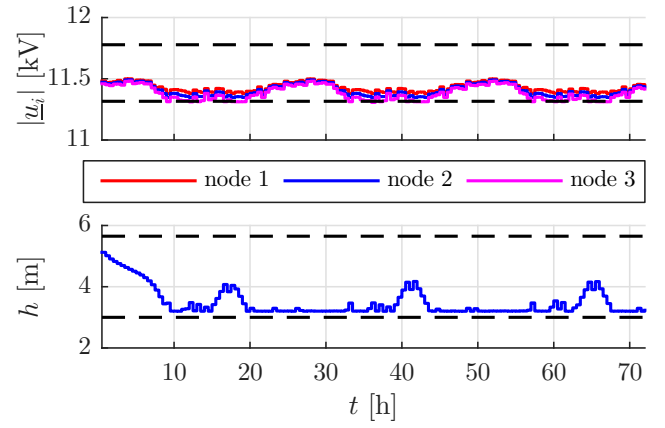


Fig. 6: Simulation results for set-up 2

However, there are some small violation at peak times due to unpredicted changes in the loads. Obviously, the controller does not exploit the full capacity of the water reservoir.

VI. CONCLUSION

In the paper, a distributed voltage controller for MV grids based on periodic DMPC is proposed. The simulation results show that the voltage can be hold in the constraints with the controller. Moreover, it is shown that the controller can be applied with adaptations to more realistic set-ups where the predictions differ from the actual values as well as the plant model does not match perfectly.

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